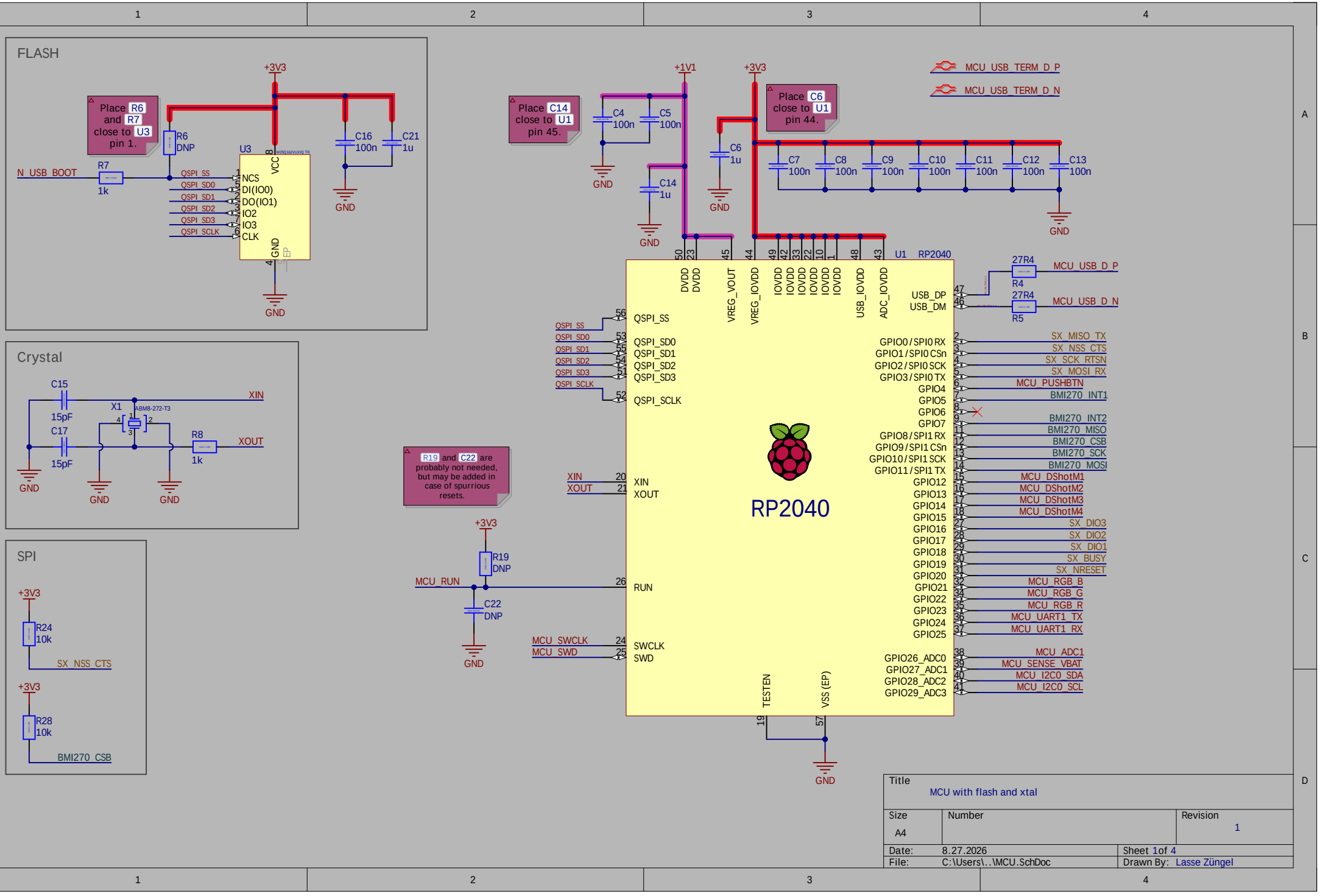
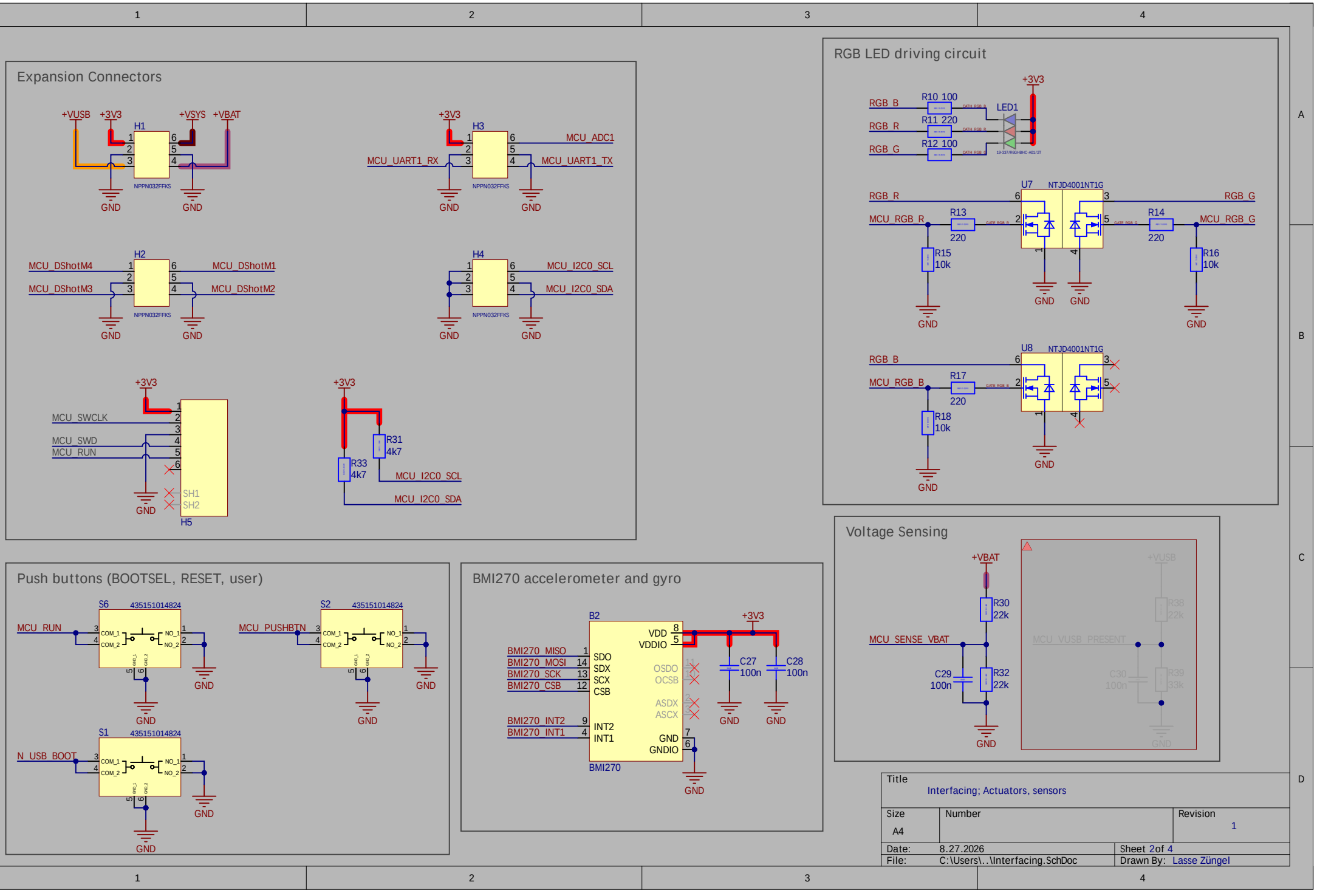
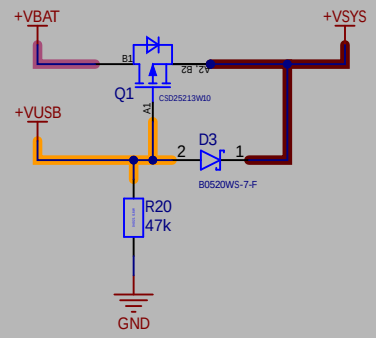
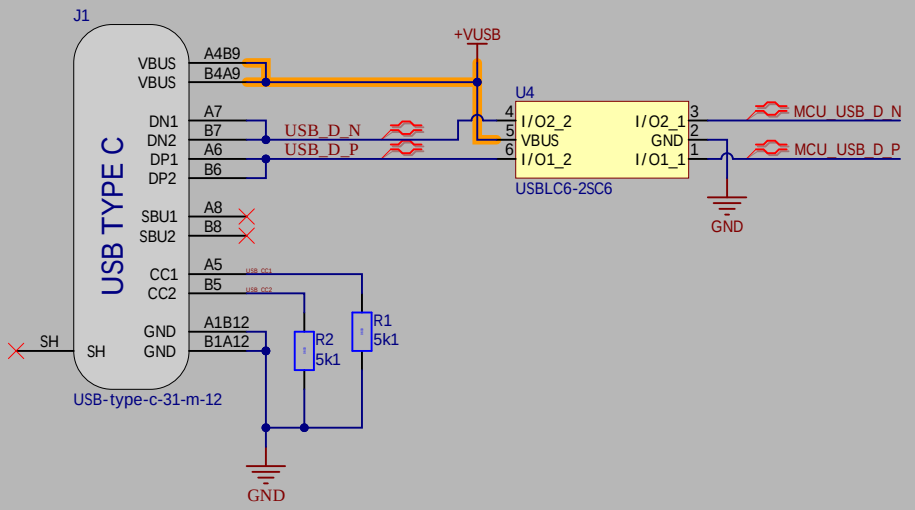

FLASH
Place R6 and R7 close to U3 pin 1.
+3V3
R6 DNP
R7 1k
N_USB_BOOT
U3
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
NCS
DI(IO0)
DO(IO1)
IO2
IO3
CLK
VCC
GND
C16 100n
C21 1u
Crystal
C15 15pF
C17 15pF
X1 ABM8-272-T3
R8 1k
XIN
XOUT
SPI
R24 10k
SX_NSS_CTS
R28 10k
BMI270_CSB
Place C14 close to U1 pin 45.
+1V1
C4 100n
C5 100n
C14 1u
Place C6 close to U1 pin 44.
C6 1u
C7 100n
C8 100n
C9 100n
C10 100n
C11 100n
C12 100n
C13 100n
MCU_USB_TERM_D_P
MCU_USB_TERM_D_N
R19 and C22 are probably not needed, but may be added in case of spurrious resets.
R19 DNP
C22 DNP
MCU_RUN
U1 RP2040
DVDD
VREG_VOUT
VREG_IOVDD
IOVDD
USB_IOVDD
ADC_IOVDD
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
XIN
XOUT
RUN
SWCLK
SWD
TESTEN
VSS (EP)
USB_DP
USB_DM
27R4
R4
R5
GPIO0 / SPI0 RX
GPIO1 / SPI0 CSn
GPIO2 / SPI0 SCK
GPIO3 / SPI0 TX
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8 / SPI1 RX
GPIO9 / SPI1 CSn
GPIO10 / SPI1 SCK
GPIO11 / SPI1 TX
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
RP2040
SX_MISO_TX
SX_NSS_CTS
SX_SCK_RTSN
SX_MOSI_RX
MCU_PUSHBTN
BMI270_INT1
BMI270_INT2
BMI270_MISO
BMI270_CSB
BMI270_SCK
BMI270_MOSI
MCU_DShotM1
MCU_DShotM2
MCU_DShotM3
MCU_DShotM4
SX_DIO3
SX_DIO2
SX_DIO1
SX_BUSY
SX_NRESET
MCU_RGB_B
MCU_RGB_G
MCU_RGB_R
MCU_UART1_TX
MCU_UART1_RX
MCU_ADC1
MCU_SENSE_VBAT
MCU_I2C0_SDA
MCU_I2C0_SCL
MCU_SWCLK
MCU_SWD
Title
MCU with flash and xtal
Size
A4
Number
Revision
1
Date: 8.27.2026
Sheet 1of 4
File: C:\Users\..\MCU.SchDoc
Drawn By: Lasse Züngel


Expansion Connectors
+VUSB
+3V3
+VSYS
+VBAT
H1
NPPN032FFKS
GND
H3
MCU_ADC1
MCU_UART1_RX
MCU_UART1_TX
H2
MCU_DShotM4
MCU_DShotM1
MCU_DShotM3
MCU_DShotM2
H4
MCU_I2C0_SCL
MCU_I2C0_SDA
MCU_SWCLK
MCU_SWD
MCU_RUN
SH1
SH2
H5
R31
4k7
R33
4k7
RGB LED driving circuit
RGB_B
RGB_R
RGB_G
R10 100
R11 220
R12 100
LED1
U7
NTJD4001NT1G
MCU_RGB_R
R13
220
R14
220
R15
10k
R16
10k
MCU_RGB_G
U8
MCU_RGB_B
R17
220
R18
10k
Push buttons (BOOTSEL, RESET, user)
S6
435151014824
S2
S1
N_USB_BOOT
BMI270 accelerometer and gyro
B2
BMI270_MISO
BMI270_MOSI
BMI270_SCK
BMI270_CSB
BMI270_INT2
BMI270_INT1
SDO
SDX
SCX
CSB
INT2
INT1
VDD
VDDIO
GND
GNDIO
OSDO
OCSB
ASDX
ASCX
C27
100n
C28
100n
BMI270
Voltage Sensing
MCU_SENSE_VBAT
R30
22k
R32
22k
C29
100n
MCU_VUSB_PRESENT
R38
22k
R39
33k
C30
100n
Title
Interfacing; Actuators, sensors
Size
A4
Number
Revision
1
Date:
8.27.2026
Sheet 2of 4
File:
C:\Users\..\Interfacing.SchDoc
Drawn By:
Lasse Züngel

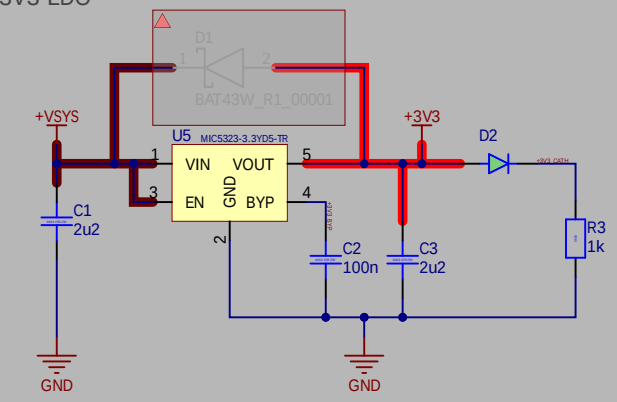
Voltage selection



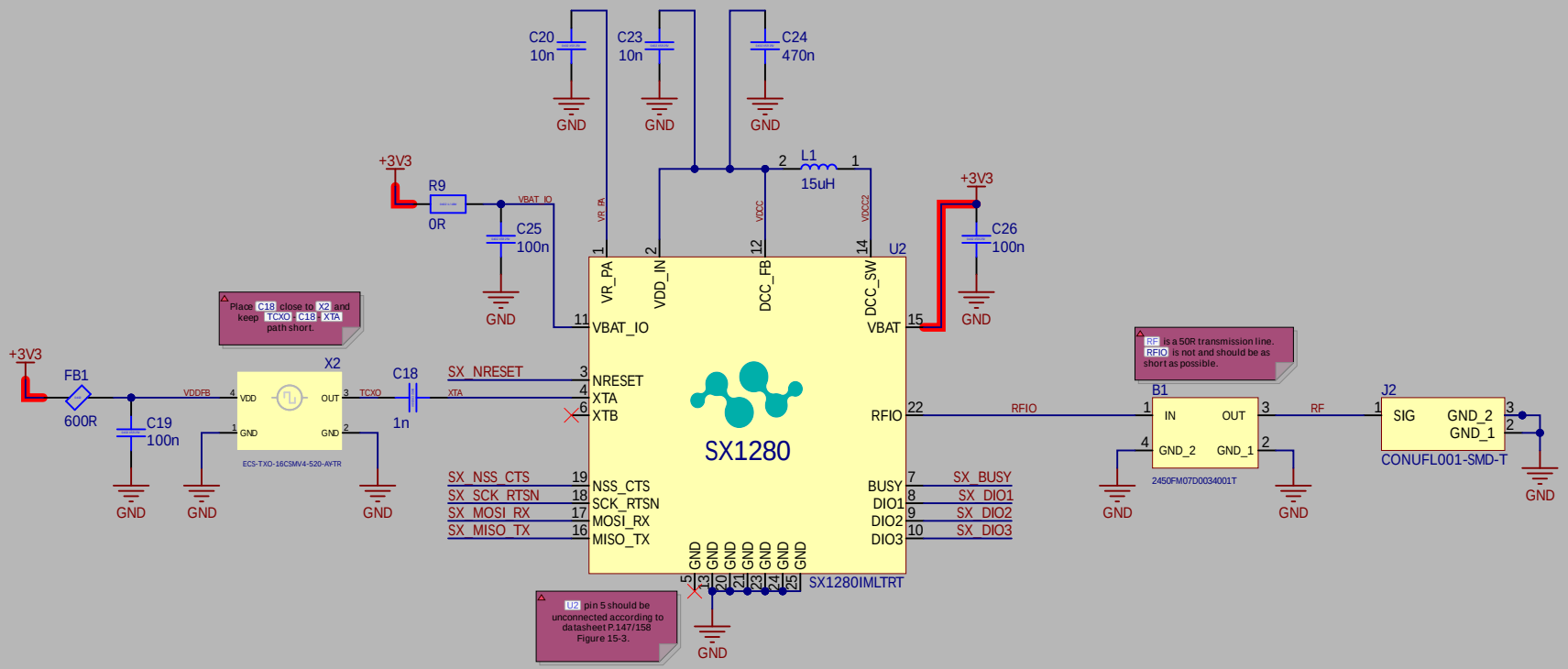
USB C



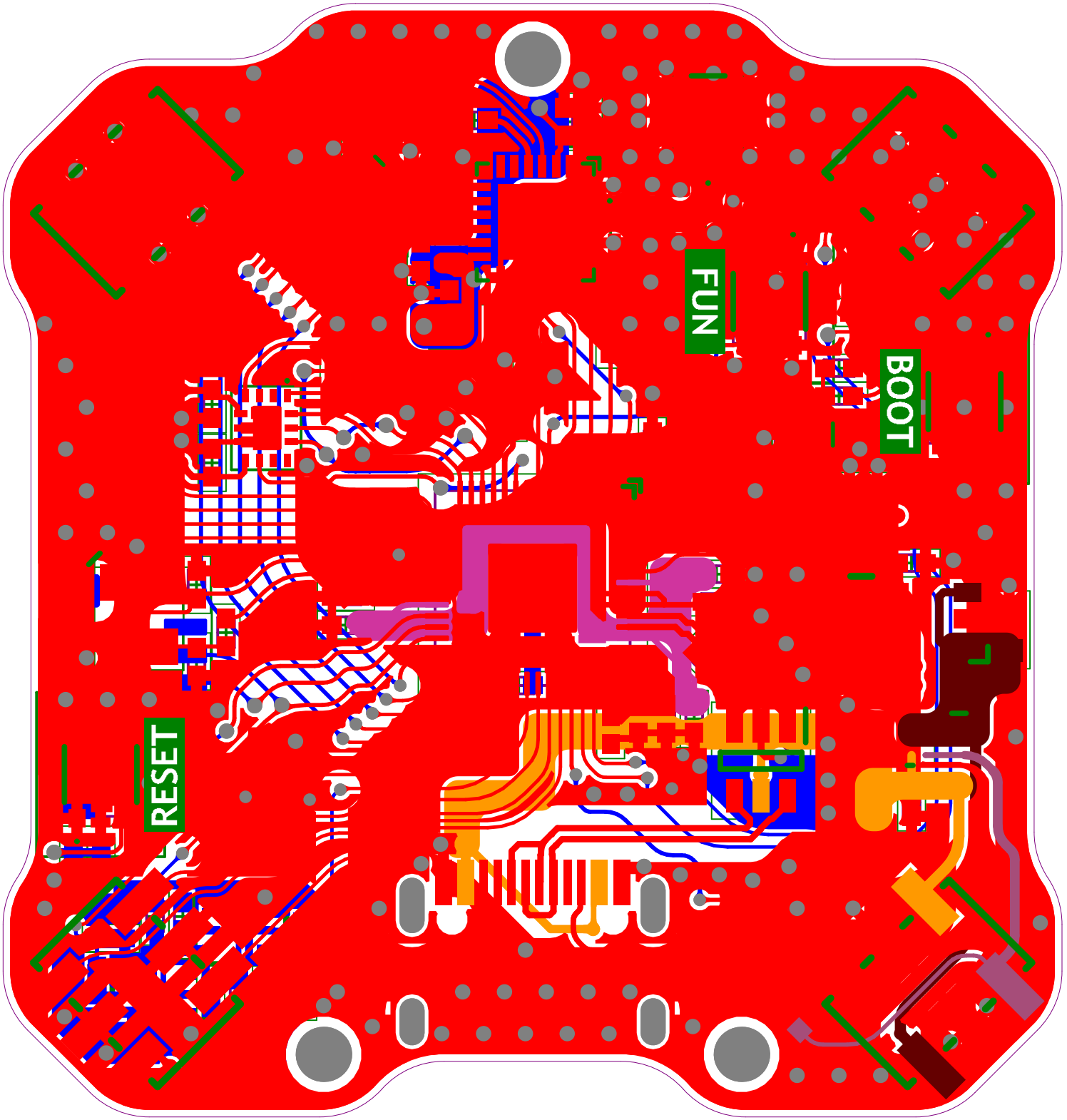
3V3 LDO



Title: Power Management and USB		
Size: A4	Number:	Revision: 1
Date: 8.27.2026		Sheet 3 of 4
File: C:\Users\..\Power.SchDoc		Drawn By: Lasse Züngel



Title: RF section		
Size: A4	Number:	Revision: 1
Date: 8.27.2026		Sheet 4 of 4
File: C:\Users\..\RF.SchDoc		Drawn By: Lasse Züngel


FUN
BOOT
RESET